

# Geohyperbolic Routing and Addressing Schemes

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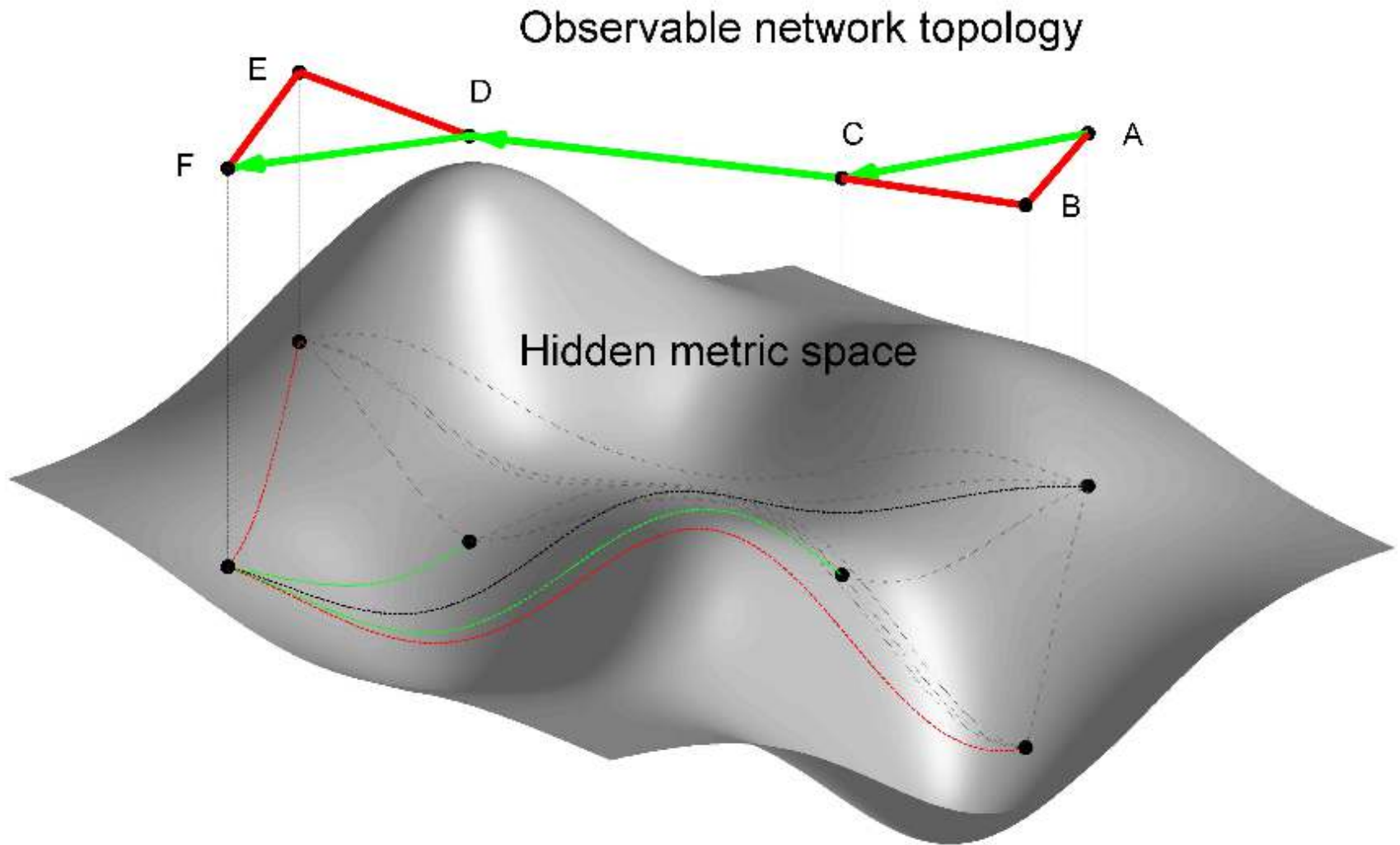
NDNcomm, March 2017



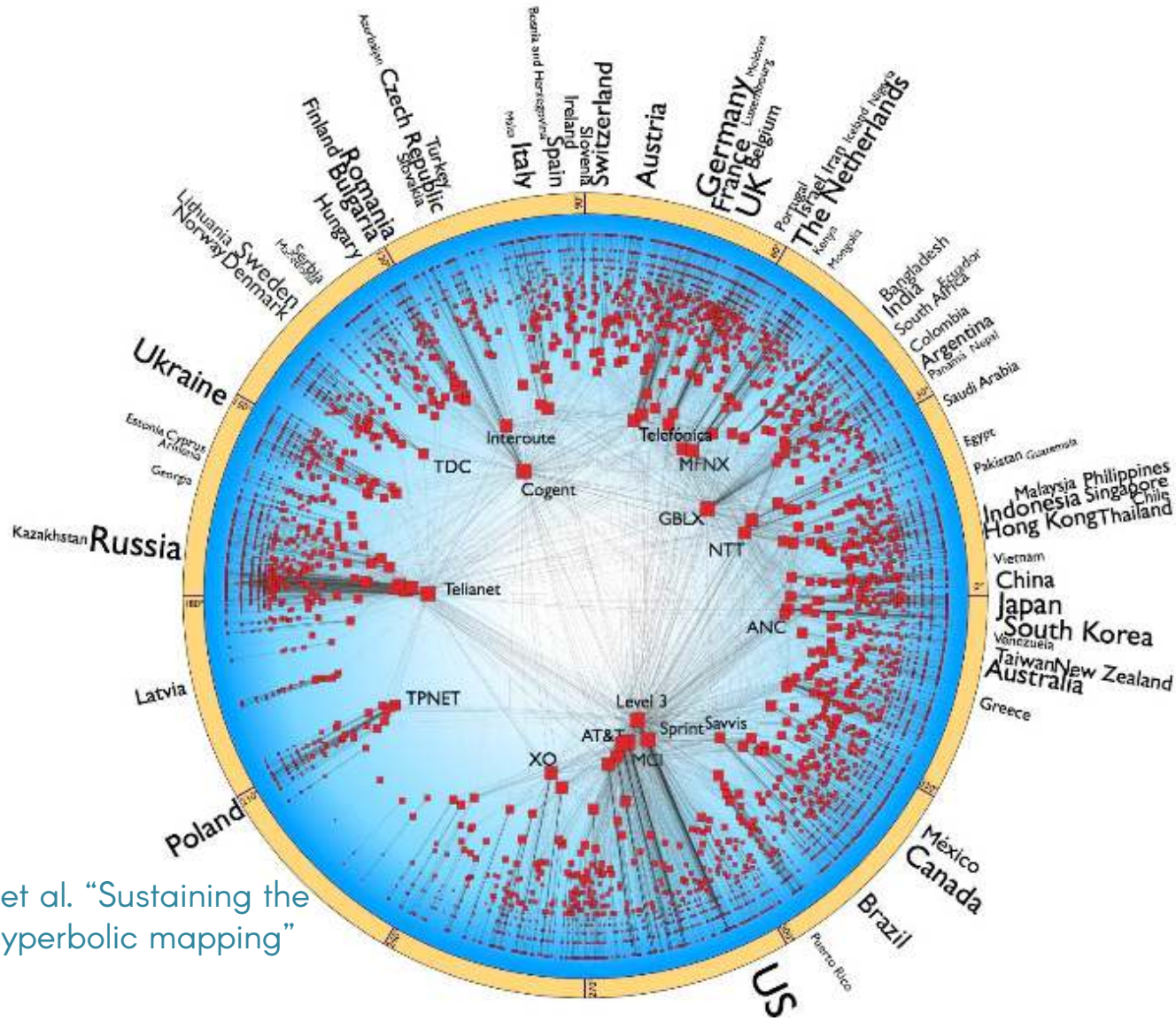
# Routing Scalability — Why Do We Care?

- Best forwarding strategy: zero routing overhead and smallest FIB size.
- BGP: linear growth of FIBs, unbounded routing overhead (persistent routing oscillations).
- Data-centric paradigms require immense addressing spaces.
- New approach to routing is needed for future networks.

# Greedy Geometric Routing

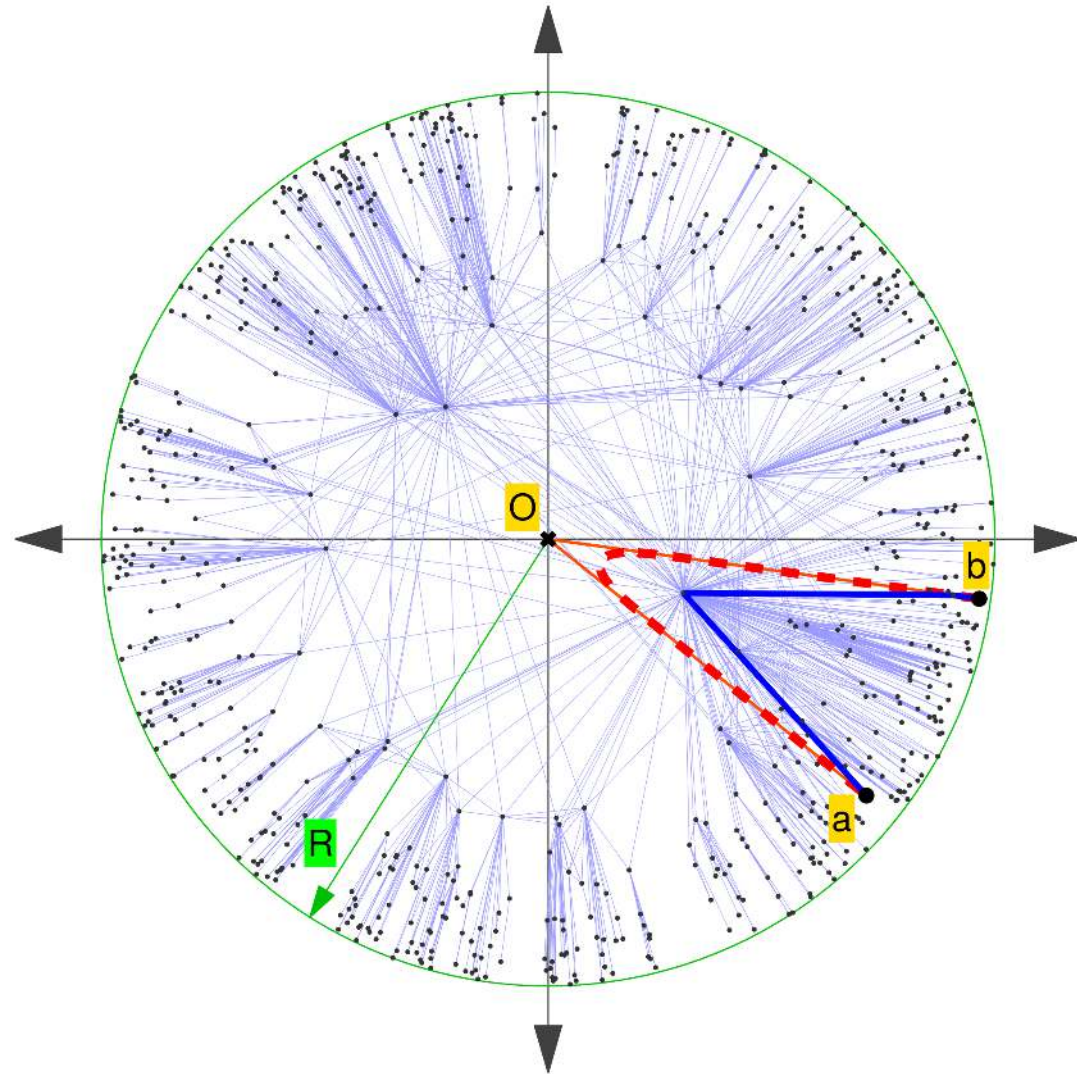


# Greedy *Hyperbolic* Routing



From Boguna et al. "Sustaining the Internet with hyperbolic mapping" (2010)

# Greedy *Hyperbolic* Routing



Geodesic lines (shortest distances) in hyperbolic space are congruent with shortest paths in network embedded in this hyperbolic space

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# Greedy *Hyperbolic* Routing

- It's not easy to make such a map: need to know substantial part of network topology; coordinate computations are not straightforward.
- Such mapping is not aware of time delays that affect user experience.

# What to do?

- Include delay information in network design: geodistance is related to round trip time (RTT).
- Map nodes to the underlying space “on-the-go”, as the network grows.
- Network should evolve in a specific manner.

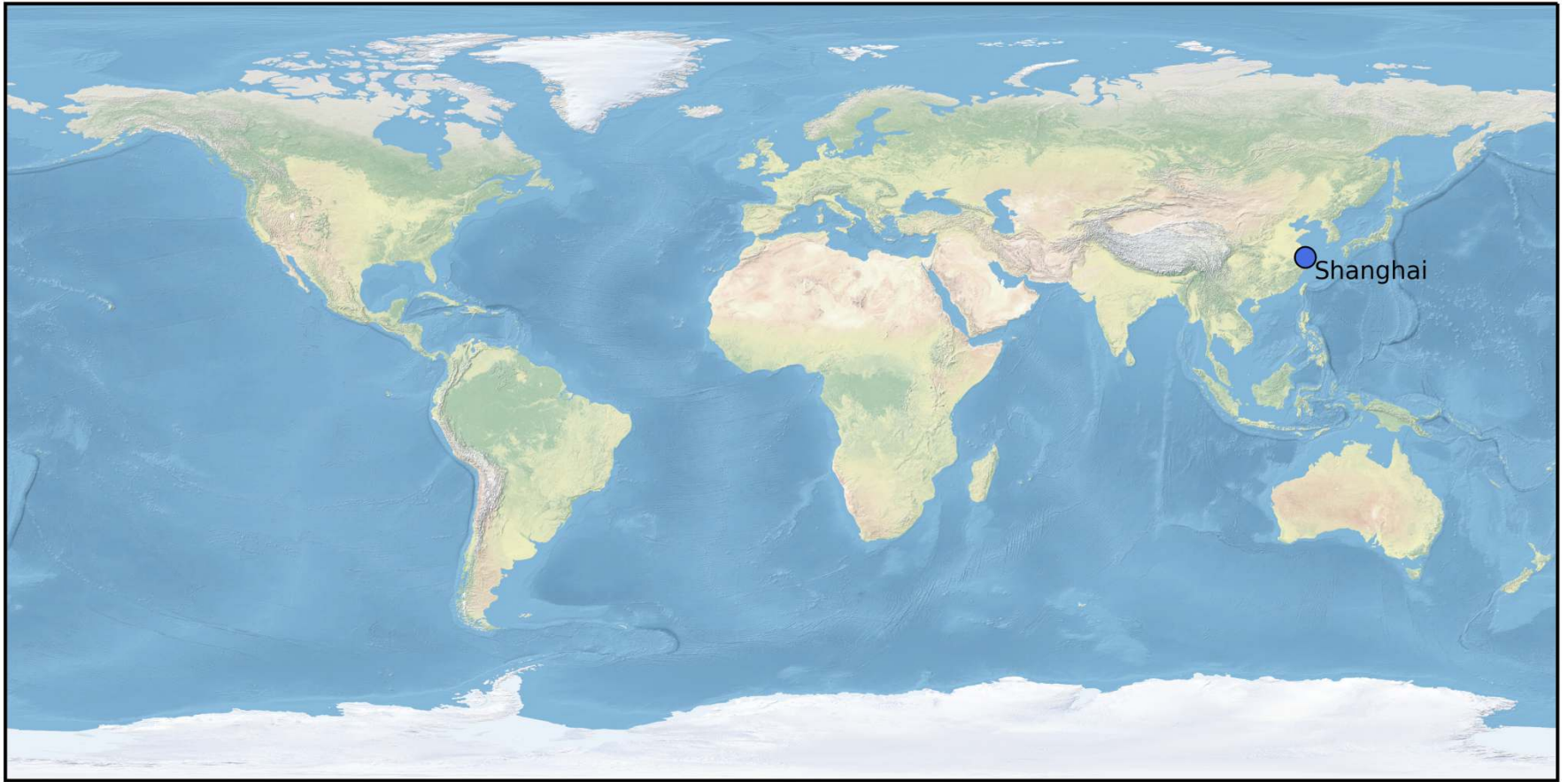
# What to do?

- General network construction scheme: a node arrives in the network and gets an address — a set of coordinates in the underlying space.
- A node connects to  $m$  geometrically closest neighbors, as defined by the distance function of the underlying space.
- Greedy routing uses the same distance function to forward packets.
- That's it!

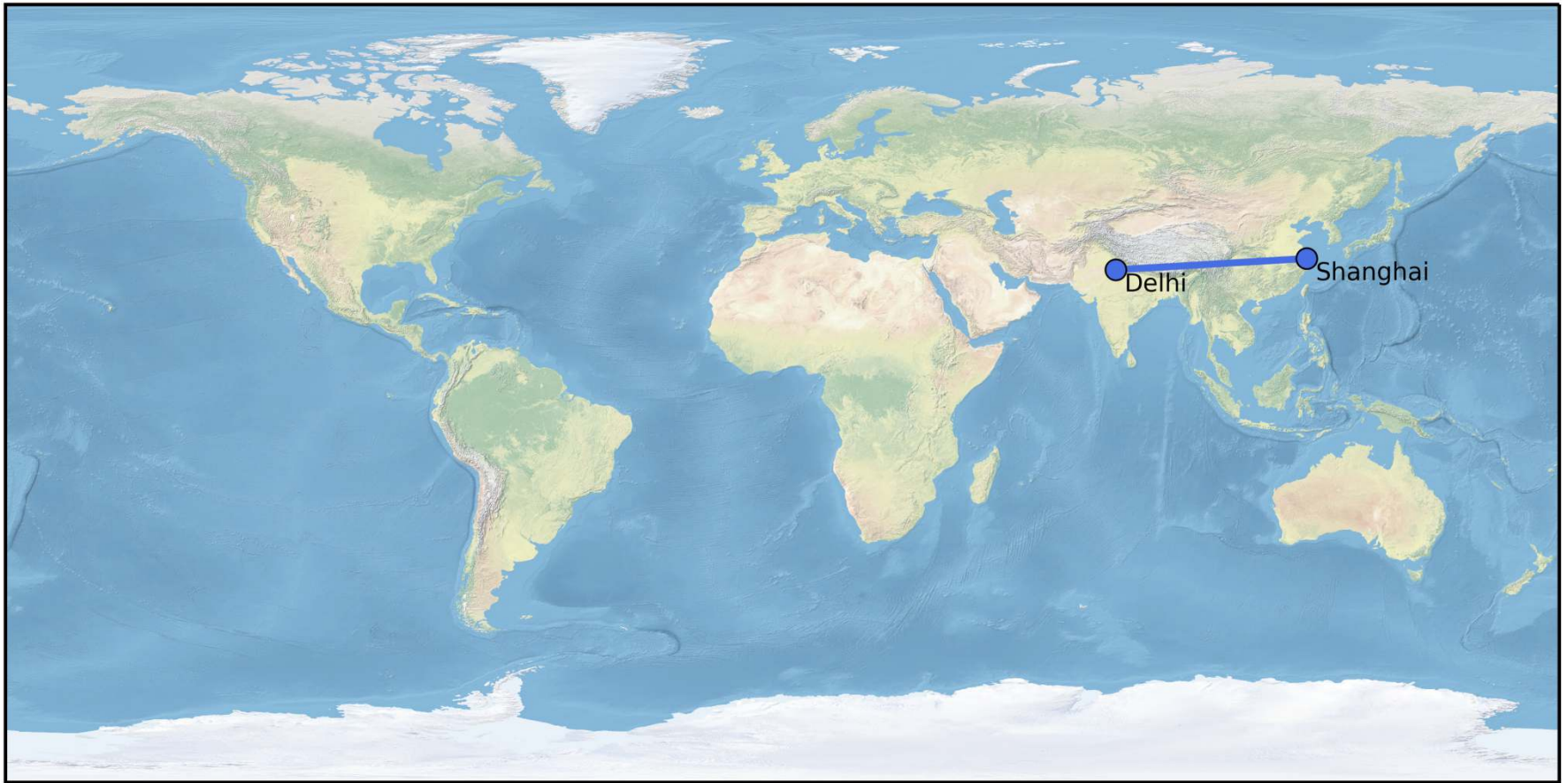
# Geographic Scheme (GEO)

- Start with the simplest model: use latitude and longitude of a node to assign an address and connect new nodes to  $m$  geographically closest neighbors.
- Resulting underlying space — two-dimensional sphere.
- Let's consider an example with  $m = 1$ .

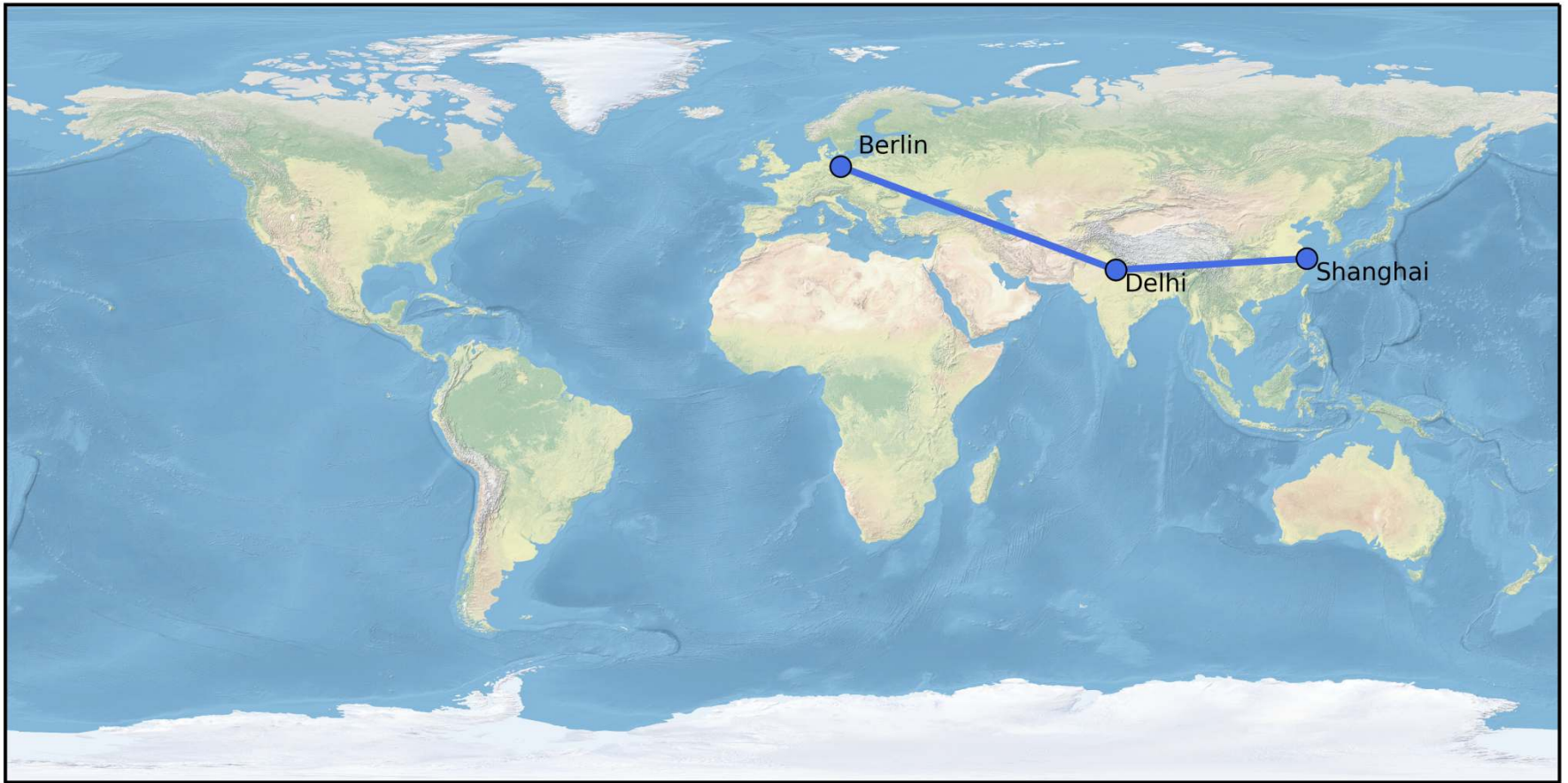
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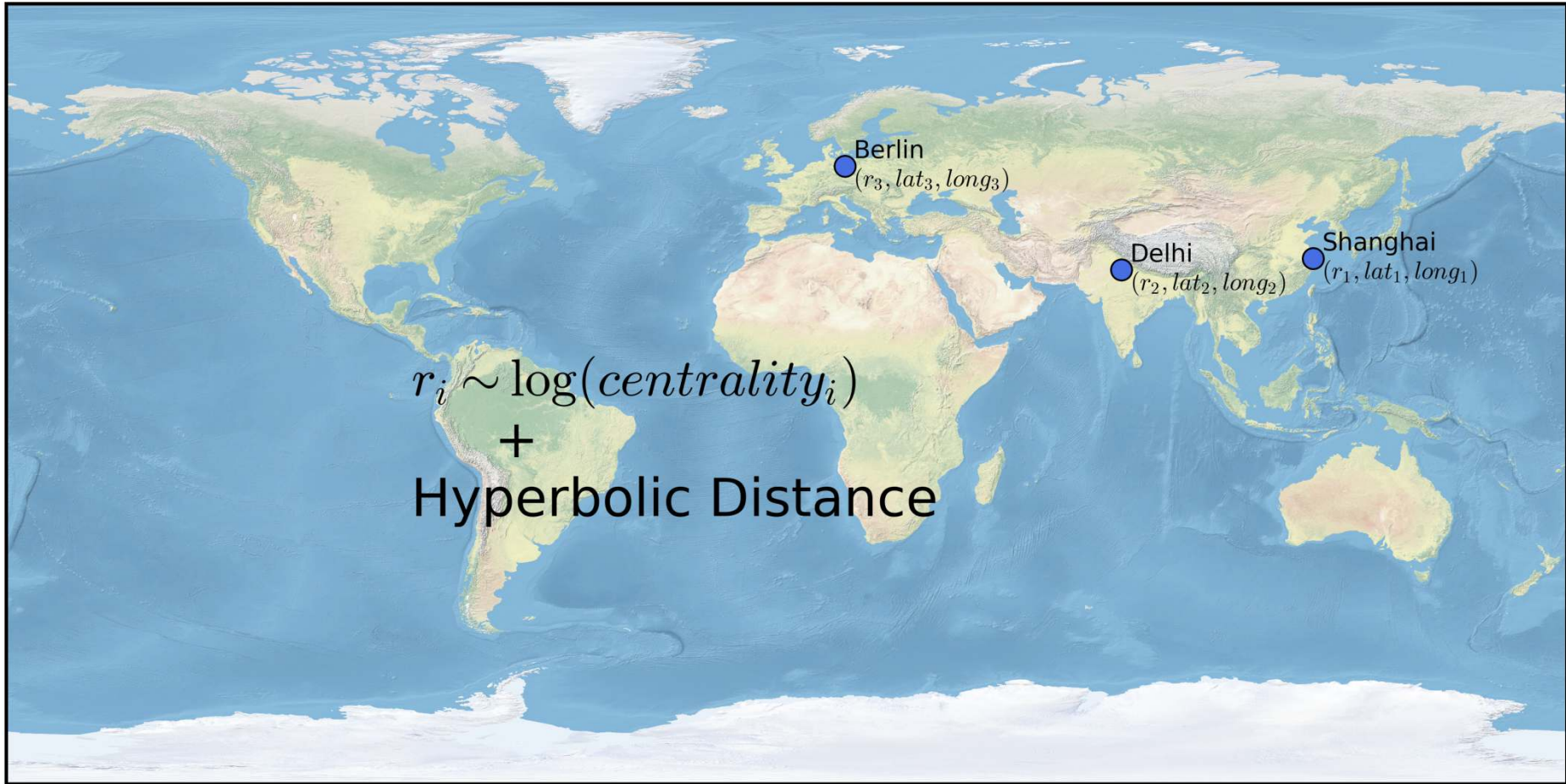
# Geographic Scheme (GEO)

- This simple scheme does not guarantee routing scalability: success ratio degrades quickly with network size, demanding for auxiliary forwarding algorithms.
- The resulting networks do not have robust topology: they resemble random geometric graphs on sphere, i.e. graphs with narrow degree distribution.

# Geohyperbolic Scheme (GH)

- Modification to the last scheme: add third, **centrality**, coordinate to a node's address that captures how “central” a node is in the network, i.e. how likely new nodes are to establish their connections with this node.
- Assign node's *radial coordinate* as a  $\sim \log(\text{centrality})$ .
- If centrality scores have heavy-tailed distribution, this assignment of coordinates naturally maps nodes to 3-dimensional *hyperbolic space*.
- Use hyperbolic distance instead of geographic distance to establish network links and perform GGR.

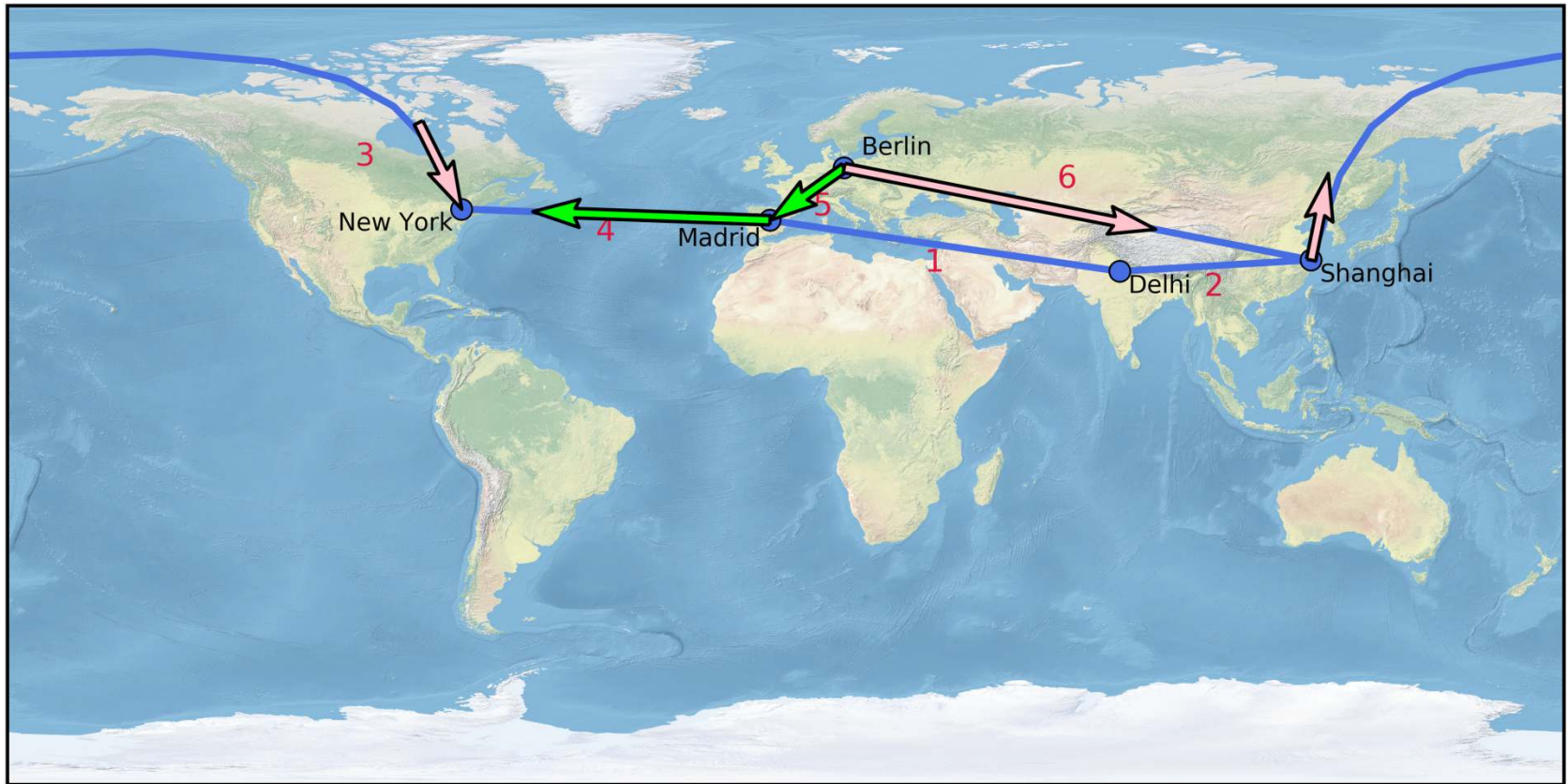
# Geohyperbolic Scheme (GH)



# Geohyperbolic Scheme (GH)

- Centrality scores can be chosen according to certain needs.
- Any geographic location is properly mapped to 3-dimensional hyperbolic space.
- Resulting networks are robust to random failures.
- Success ratio of GGR is almost 1 for any network size and under severe connectivity failures.
- However, suboptimal delay-wise performance is observed.

# Geohyperbolic Scheme (GH)

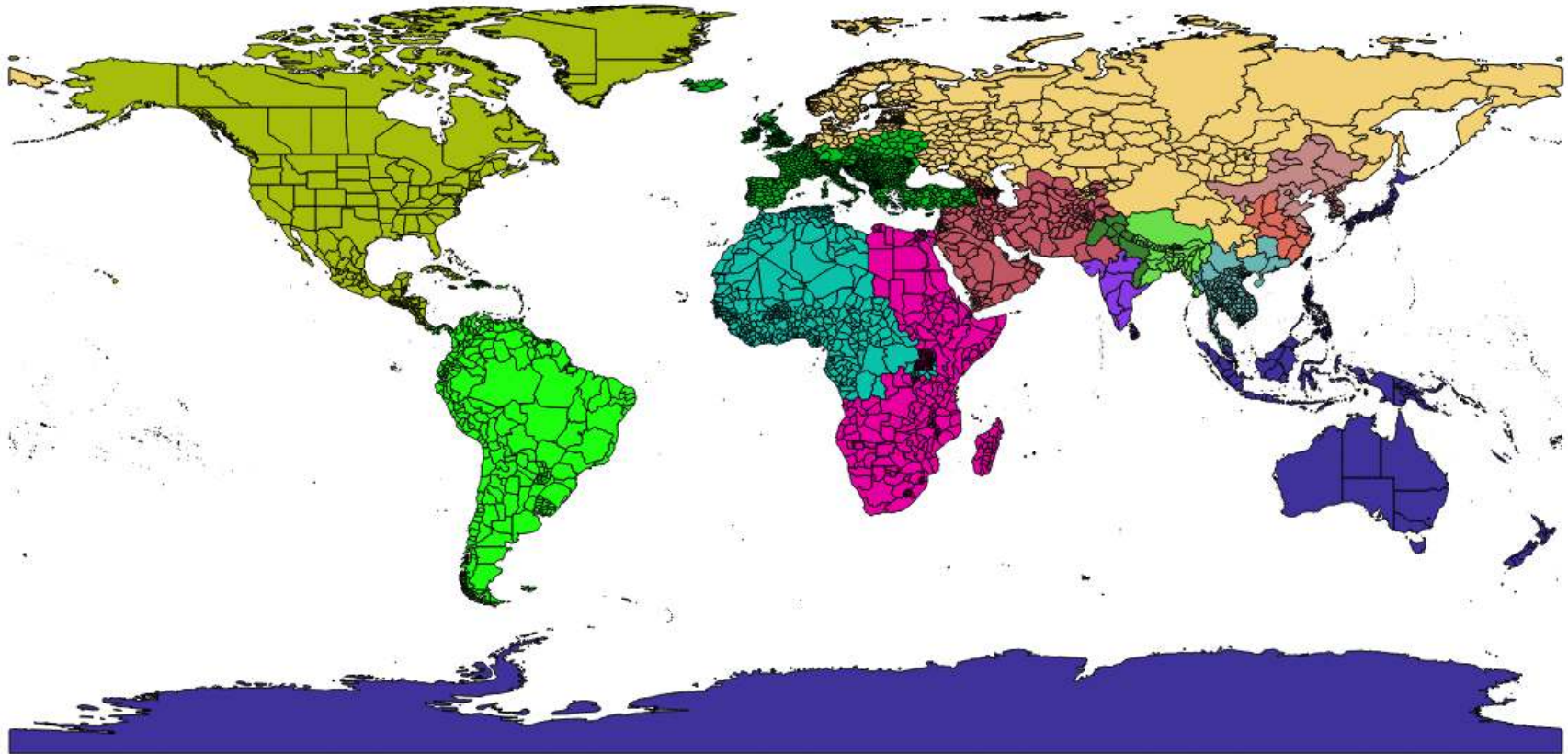


- Bad delay performance example: packet forwarding from Berlin to New York via “super central” node in Shanghai.

# Regionalized Geohyperbolic Scheme (RGH)

- Small tweak of a previous scheme: place multiple “local hubs” within large geographic regions to “attract” greedily forwarded packets from peripheral local nodes.
- Local hubs have the same radial coordinates, i.e., greedily forwarded packets are attracted to *geographically closest* local hubs, which reduces long-delay paths.

# Regionalized Geohyperbolic Scheme (RGH)



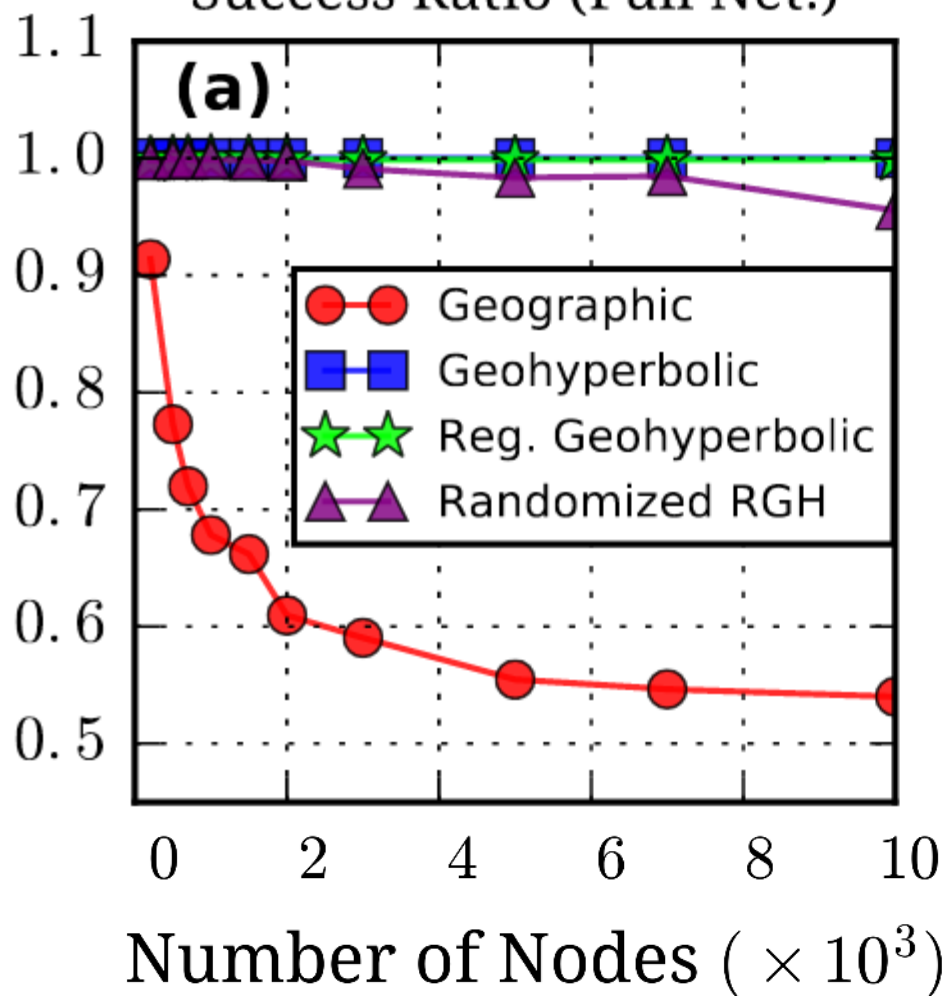
Administrative Level 1 Units (e.g., states or provinces) are merged into large regions equipped with local hubs.

# Validation

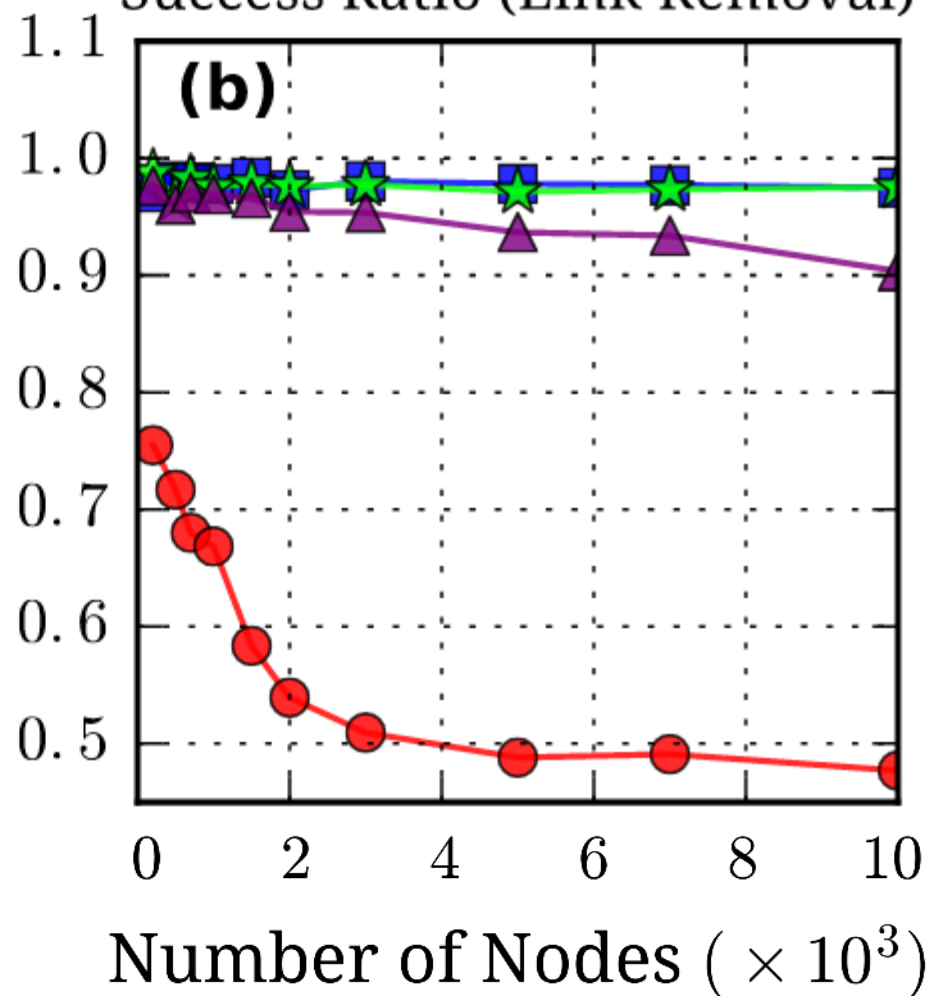
- Tested GEO, GH and RGH schemes both in large synthetic networks and NDN testbed simulations.
- Nodes appearing in cities, towns, etc.
- Centrality score of a node is set to population of corresponding populated place:  
$$r \sim \log(\text{population rank})$$
- Randomized order of nodes' arrivals in RGH scheme mimics real-world situation: central nodes are more likely to appear in densely populated places.
- Tested connectivity disruptions: 20% of links were randomly removed and greedy routing is tested again.

# Validation

Success Ratio (Full Net.)



Success Ratio (Link Removal)



# Delay Performance Metrics

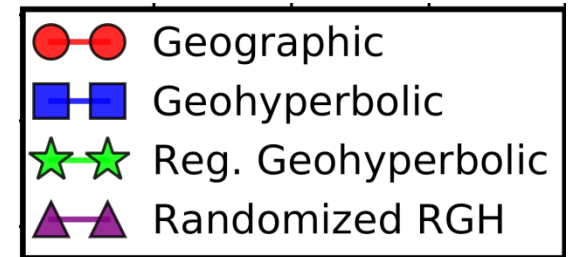
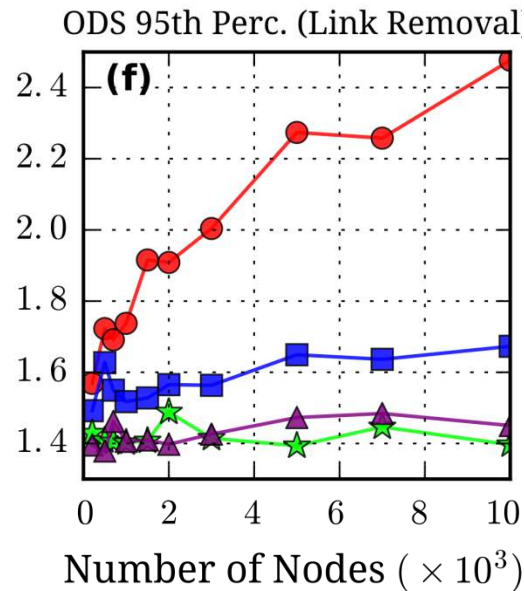
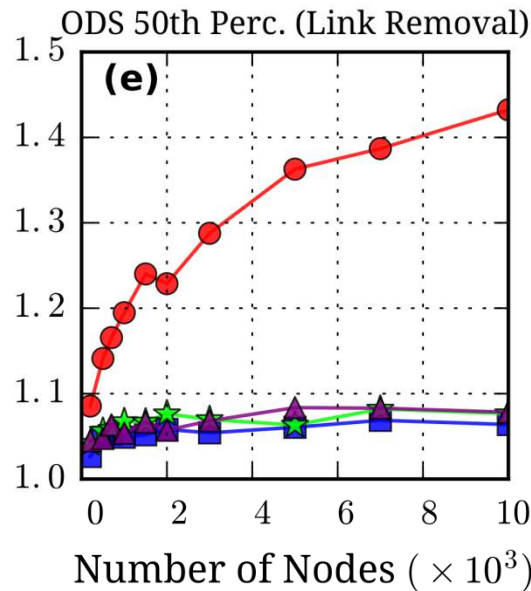
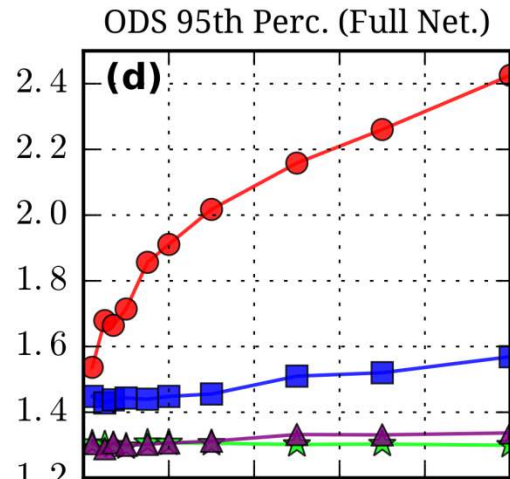
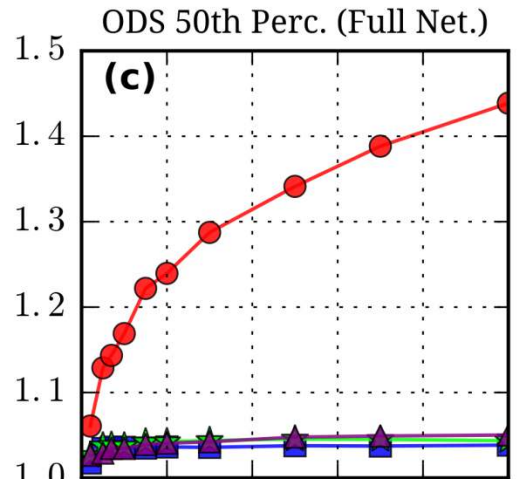
Underlay delay stretch (UDS):

$$\frac{\text{greedy path delay}}{\text{direct (underlay) delay}}$$

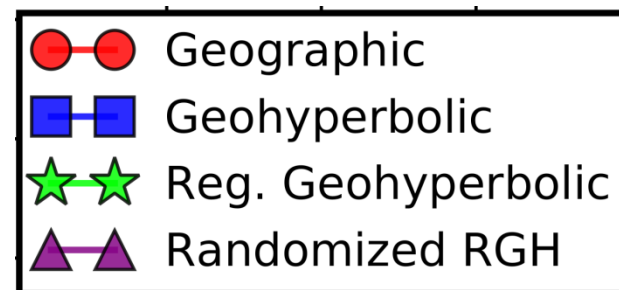
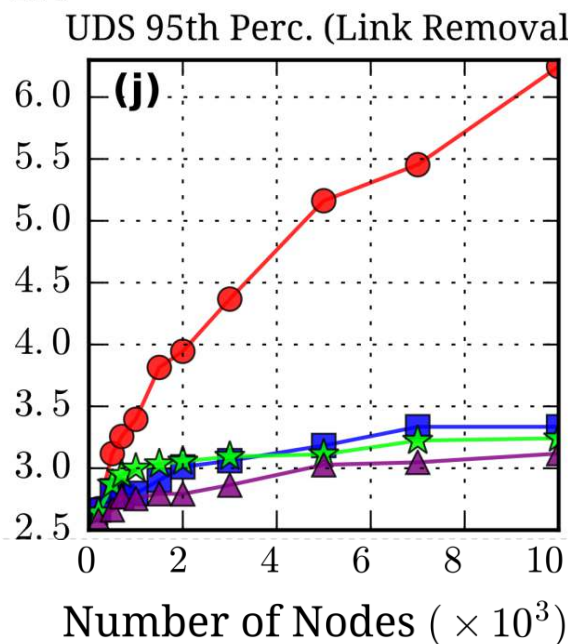
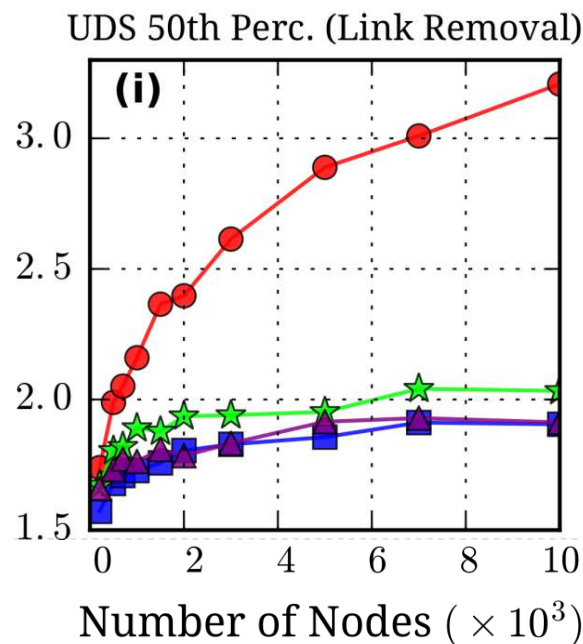
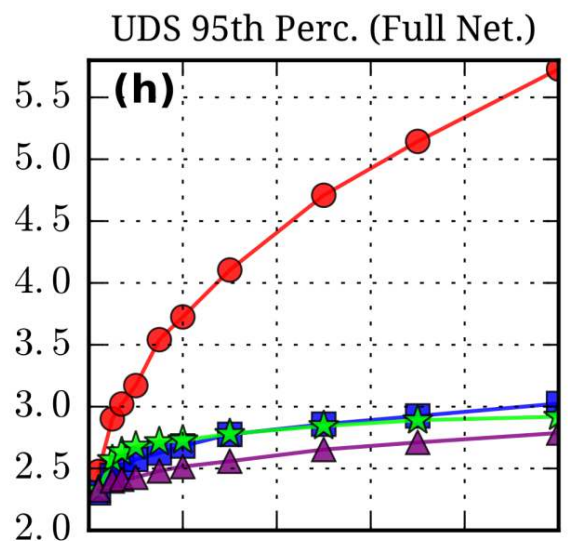
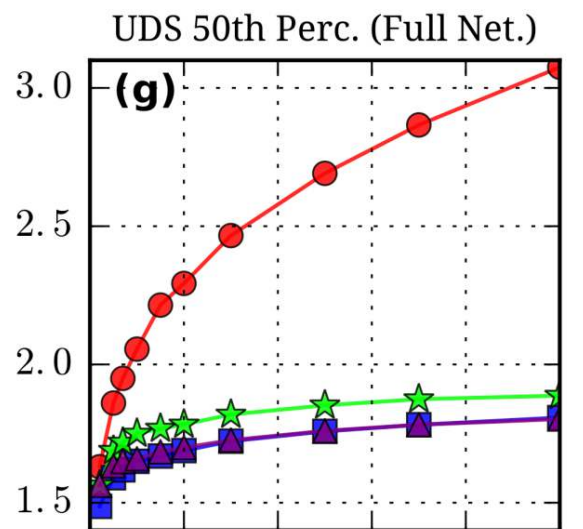
Overlay delay stretch (ODS):

$$\frac{\text{Greedy path delay}}{\text{Dijkstra path overlay delay}}$$

# Validation



# Validation



# Validation

- NDN testbed: used 29 functional nodes and links according to three schemes.
- Best SR performance: GH and RGH.
- Under 20% links damage SR is still high: 0.98 and 0.97.
- Median ODS: 1 in all cases.
- 95-th percentile ODS: 1.54 for RGH, 2.65 for GH (regionalization helps!)

# Conclusion

- GEO scheme does not offer scalable solution.
- SR is almost independent of network size for GH and RGH.
- SR is high even under severe connectivity failures.
- Delay stretch is bounded.
- Only small fraction of paths should be optimized via auxiliary forwarding algorithms!

# Conclusion

- Proposed network design offers light-weight and scalable routing solution.
- Only geolocation of nodes are used: no virtual coordinates now!
- Flexible centrality scheme allows to implement such networks in different circumstances.
- Future work: mobile nodes?

Thank you!